BIM488 Introduction to Pattern Recognition

Feature Selection

Outline

- Introduction
- Feature Selection Methods
- Exhaustive Search
- SBS/SFS
- GSFS/GSBS
- PTA

Introduction

- Feature selection is an essential topic in the field of pattern recognition.
- The feature selection strategy has a direct influence on the accuracy and processing time of pattern recognition applications.

Introduction

Prior to any feature selection, data preprocessing is a necessary step.

- Data Preprocessing
  - Outlier removal: An outlier is defined as a point that lies very far from the mean of the corresponding random variable. Such points result in large errors during training. If such points are the result of erroneous measurements, they have to be removed.
  - Data normalization: Features with large values have large influence compared to others with small values, although this may not necessarily reflect a respective significance towards the design of the classifier.

Introduction

- A common technique is to normalize each feature via the respective estimate of the mean and variance (i.e., zero mean, unit variance. That is for the k^{th} feature:

$$\bar{x}_k = \frac{1}{N}\sum_{i=1}^{N} x_{ik} \quad , \; k = 1,2,\ldots,l$$

$$\sigma_k^2 = \frac{1}{N-1}\sum_{i=1}^{N} (x_{ik} - \bar{x}_k)^2$$

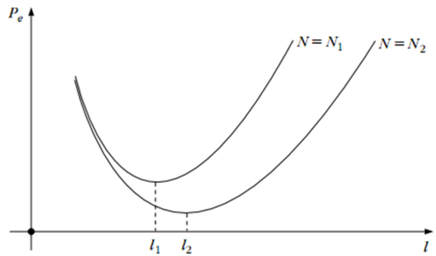
$$\hat{x}_{ik} = \frac{x_{ik} - \bar{x}_k}{\sigma_k}$$

Introduction

- The Peaking Phenomenon

  If, in an ideal world, the class pdfs were known, then increasing the number of features would be beneficial.

  In practice, the general trend is that for a finite number of training points, increasing the number of features initially improves the generalization error rate, but after a certain value, the generalization error rate increases.

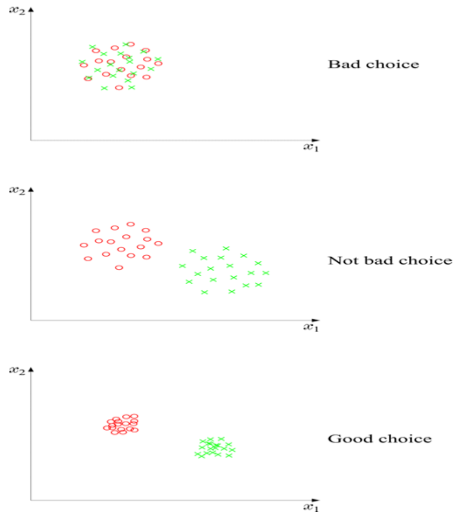


Introduction

- The main goals in feature selection:
  - Select the "optimum" number l of features
  - Select the "best" l features

- Large l has a three-fold disadvantage:
  - High computational demands
  - Low generalization performance
  - Poor error estimates

Introduction



Feature Selection Methods

Widely used feature selection methods are:

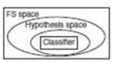
- **Filters:** ranks features independently of the classifier.
- **Wrapper:** employs a classifier to assess feature subsets.

- *Univariate approach*: considers one feature at a time.
- *Multivariate approach:* considers subsets of features together.

Feature Selection Methods

Model search	Advantages	Disadvantages	Examples
Filter	Univariate		
	Fast Scalable Independent of the classifier	Ignores feature dependencies Ignores interaction with the classifier	χ^2 Euclidean distance *i*-test Information gain, Gain ratio (Ben-Bassat, 1982)
	Multivariate		
	Models feature dependencies Independent of the classifier Better computational complexity than wrapper methods	Slower than univariate techniques Less scalable than univariate techniques Ignores interaction with the classifier	Correlation-based feature selection (CFS) (Hall, 1999) Markov blanket filter (MBF) (Koller and Sahami, 1996) Fast correlation-based feature selection (FCBF) (Yu and Liu, 2004)

Feature Selection Methods

Wrapper	Deterministic		
	Simple Interacts with the classifier Models feature dependencies Less computationally intensive than randomized methods	Risk of over fitting More prone than randomized algorithms to getting stuck in a local optimum (greedy search) Classifier dependent selection	Sequential forward selection (SFS) (Kittler, 1978) Sequential backward elimination (SBE) (Kittler, 1978) Plus q take-away r (Ferri *et al.*, 1994) Beam search (Siedelecky and Sklansky, 1988)
	Randomized		
	Less prone to local optima Interacts with the classifier Models feature dependencies	Computationally intensive Classifier dependent selection Higher risk of overfitting than deterministic algorithms	Simulated annealing Randomized hill climbing (Skalak, 1994) Genetic algorithms (Holland, 1975) Estimation of distribution algorithms (Inza *et al.*, 2000)

Feature Selection Methods

- In this course, we are going to learn some of the well-known wrapper methods including:
  - Exhaustive Search
  - SBS/SFS
  - GSFS/GSBS
  - PTA

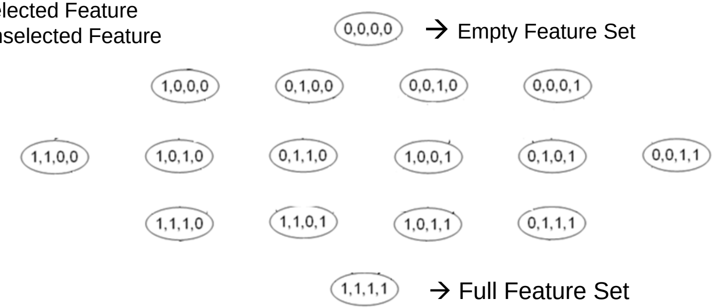
Exhaustive Search

- In this selection method, (*N, d*) possible feature combinations are analyzed to obtain optimal *d* dimensional feature subset out of *N* dimensional full feature set based on a criterion function (e.g. **classification accuracy**).
- Although this method guarantees to reach the optimal solution, required processing time is quite high even for moderate number of features.

Exhaustive Search

- Example: 4-dimensional feature set

1: Selected Feature
0: Unselected Feature



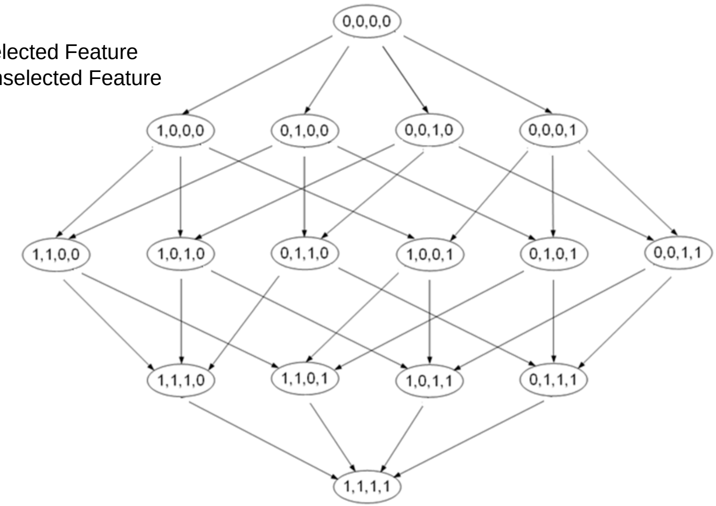
- How many feature combinations for N-dimensional feature set?

Sequential forward selection (SFS)

- It operates in bottom-to-top manner.
- The selection procedure starts with an empty set initially.
- Then, at each step, the feature maximizing the criterion function is added to the current set.
- This operation continues until the desired number of features is selected.
- The nesting effect is present such that a feature added into the set in a step can not be removed in the subsequent steps.
- As a consequence, SFS method can offer only suboptimal result.

Sequential forward selection (SFS)

1: Selected Feature
0: Unselected Feature

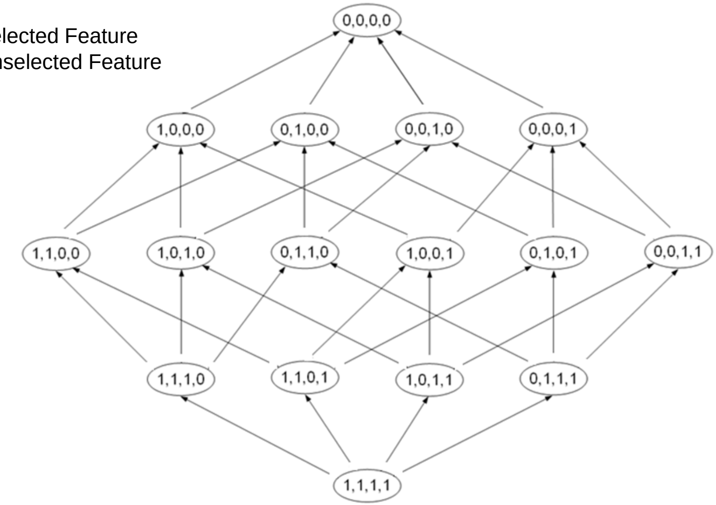


Sequential backward selection (SBS)

- SBS works in a top-to-bottom manner.
- It is the reverse case of SFS method.
- Initially, complete feature set is considered. At each step, single feature is removed from the current set so that the criterion function is maximized for the remaining features within the set.
- Removal operation continues until the desired number of features is obtained.
- The nesting effect is present in this method as in SFS. Once a feature is eliminated from the set, it can not enter into the set in the subsequent steps.
- Thus, SBS offers suboptimal solution.

Sequential backward selection (SBS)

1: Selected Feature
0: Unselected Feature



Generalized SFS

- In generalized version of SFS, instead of single feature, *n* features are added to the current feature set at each step.
- The nesting effect is still present.

Generalized SBS

- In generalized form of SBS (GSBS), instead of single feature, *n* features are removed from the current feature set at each step.
- The nesting effect is present here, too.

Plus-*l* takeaway-*r* (PTA)

- The nesting effect present in SFS and SBS can be partly avoided by moving in the reverse direction of selection for certain number of steps.
- With this purpose, at each step, *l* features are selected using SFS and then *r* features are removed with SBS.
- This method is called as PTA.
- Although the nesting effect is reduced with respect to SFS and SBS, PTA still provides suboptimal results.

Summary

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- Feature Selection Methods
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References

- S. Theodoridis and K. Koutroumbas, *Pattern Recognition* (4th Edition), Academic Press, 2009.
- Saeys Y., Inza I., Larranaga P., "A review of feature selection techniques in bioinformatics", Bioinformatics, 23(19), 2507-2517, 2007.